



CHAPTER ATA 80 ENGINE STARTING

THIS MANUAL IS INTENDED FOR TRAINING PURPOSE ONLY

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Introduction

The engine start system provides the means to start the engines on the ground and provide re-starts in the air.

General Description

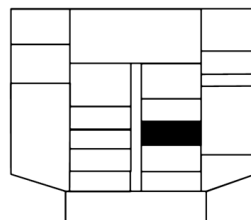
The engine control system is powered up when the appropriate airframe 28 Vdc Essential Bus is selected ON. Upon being powered up the Full Authority Digital Electronic Control (FADEC) performs a series of initialization tests on

Starting

internal functions and interfaces and begins transmitting data on ARINC 429 and RS 422 serial data buses.

[Refer Figure 80-1](#)

Select the ignition switch to the NORMAL position. An engine start is then initiated by selecting the engine and activating the starter by pressing the engine START switch and moving the Condition Lever from the FUEL OFF, to START FEATHER position.



OVERHEAD CONSOLE

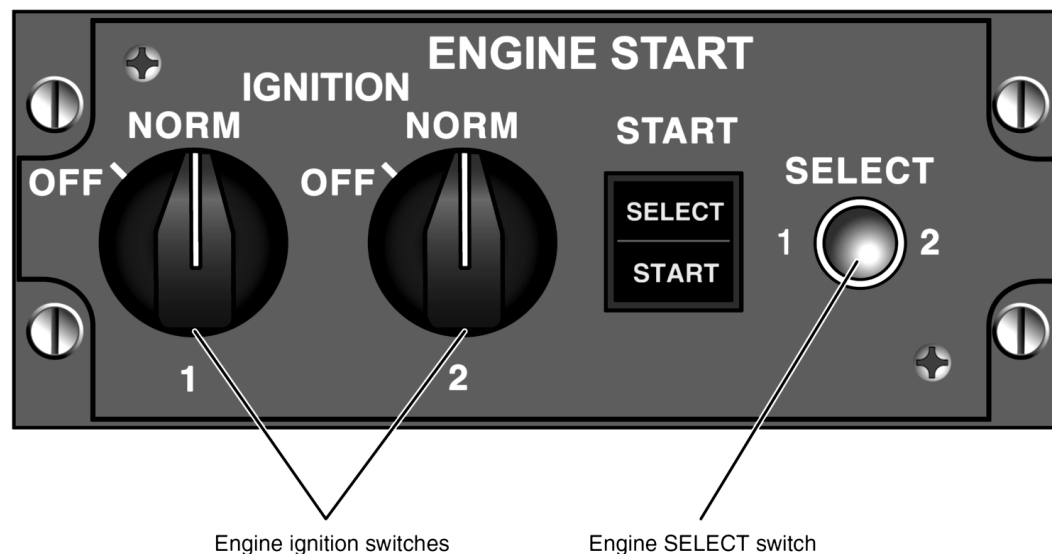


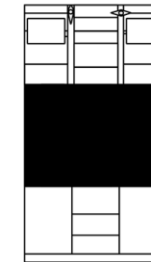
Figure 80-1 Engine Start Control Panel

Detailed Description

[Refer Figures 80-2 & 80-3](#)

The movement of the Condition Lever out of the “FUEL OFF” position causes the following to occur:

- Condition Lever mounted Start/Shutdown microswitches transition from the closed circuit to the open circuit state (one per channel) resetting the independent N_h Overspeed (O/S) protection circuitry to unlatch any previous O/S trip.
- Wrap-around signals from the Start/Shutdown discretes are input to the FADEC controller, signaling to the software that a start has been commanded.
- When N_h is greater than 8% and Condition Lever Angle (CLA) is out of the “FUEL OFF” position, the FADEC begins metering fuel flow with the metering valve. **If in flight, the FADEC deenergizes the Fuel Metering Unit (FMU) O/S solenoid. On the ground, the FMU O/S solenoid is normally deenergized (open) when the engine is shutdown unless the FMU is determined to be faulty.**



CENTER CONSOLE

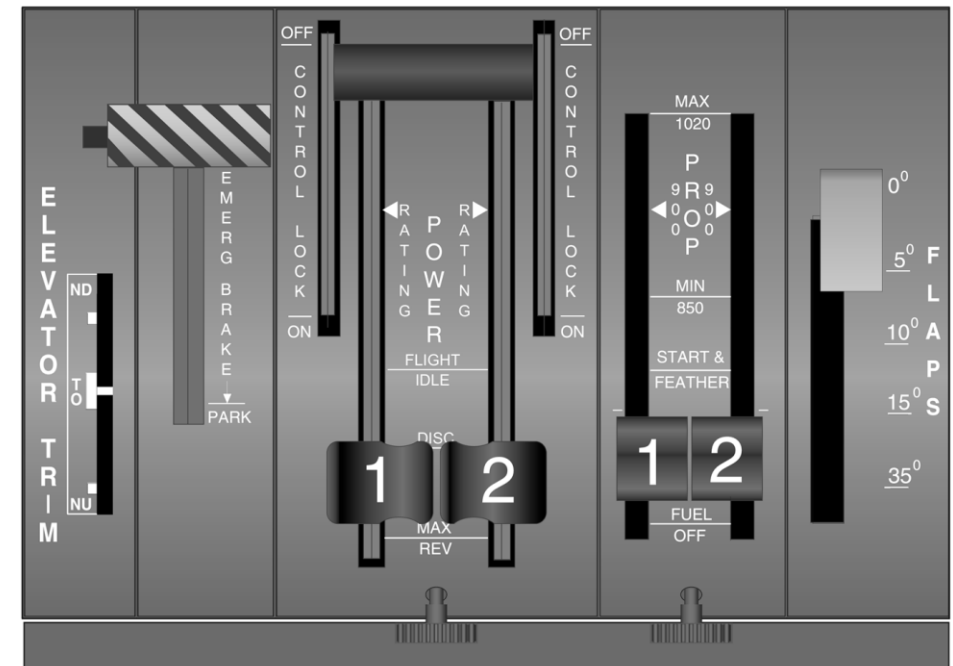


Figure 80-2 Power and Condition Levers

TYPE	RVDT	SWITCH	SWITCH	SWITCH	RVDT	SWITCH
DESIGNATION	0° TO 95° CLA	LEFT 15° CLA ENGINE START/SHUTDOWN	LEFT 15° CLA ENGINE START/SHUTDOWN	LEFT 15° CLA HYD EDP OIL FLAP DOOR	0° TO 95° LEFT CLA	LEFT 40° CLA ALTERNATE FEATHER LOCKOUT
TYPE		CLA AT 15°	CLA AT 15°	CLA AT 15°		CLA AT 40°
INTERFACE	PEC LEFT CHANNEL A	FADEC LEFT CHANNEL A	FADEC LEFT CHANNEL B	#1 ENG HYD PUMP CAUTION OIL FLAP DOOR TEST	PEC LEFT CHANNEL B	PEC + MANUAL UNFEATHER SWITCH + TIME DELAY RELAY ALTERNATE FEATHER SWITCH

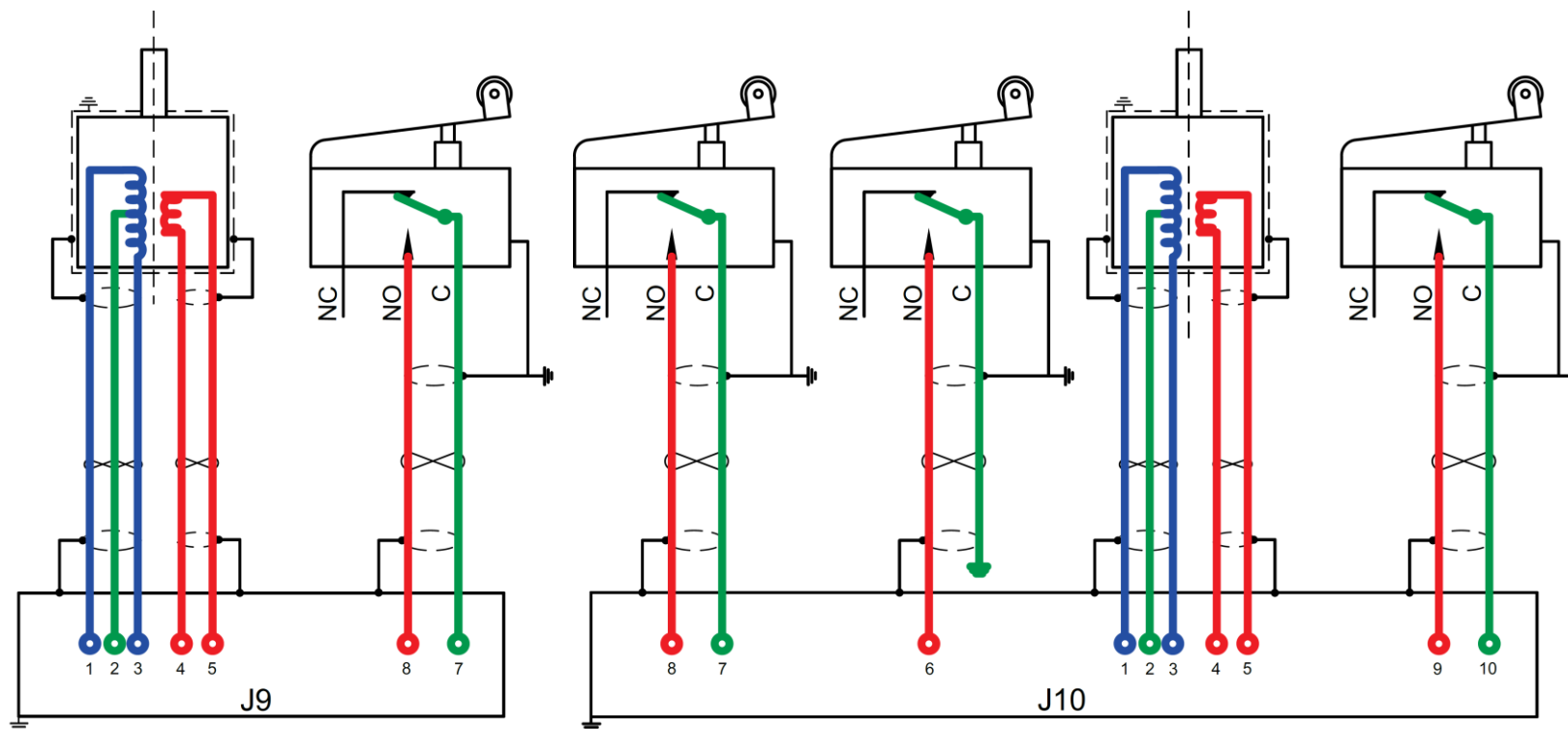


Figure 80-3 Condition Lever Microswitches (MIN to MAX)

Automatic Engine Starts on the Ground

When the start has been successfully initiated, the FADEC controls the starting sequence in the manner that follows:

- When the starter has increased the generator speed (Nh) to 8%, the FADEC commands ignition ON and schedules fuel flow as a function of Nh, ambient conditions (Temperature and Pressure), and Main Oil Temperature (MOT).
- Only one of the two ignitors is turned on. If the engine does not light within 8 seconds of fuel flow on, the FADEC turns on both ignitors and logs a fault against the faulty ignitor. Light off of engine is defined as an increase of 20 °C in Indicated Turbine Temperature (ITT)
- During run up to idle, the FADEC switches channels to test the health of the opposite channel.
- When Nh is greater than 50%, the ignitor(s) is automatically turned off
- FADEC controls engine run-up to the commanded Nh idle speed.

During ground starts, to ensure that the engine start does not cause over temperature, the FADEC has active Indicated Turbine Temperature (ITT) limiting. FADEC will reduce the fuel flow below the standard start schedule if necessary to prevent an over temperature.

FADEC will automatically abort the start, and shutdown the engine if any one of the following conditions occur:

- The engine does not light within 16 seconds of fuel flow on. FADEC shuts down the engine and logs the appropriate fault
- ITT limit of 920 °C is exceeded
- NH does not reach 50% within 70 seconds (i.e. hung or slow start).

Automatic Starts in Flight

In-flight starts are similar to ground starts except for the following:

- Both ignitors are commanded ON during the start
- The auto abort features are disabled
- FADEC does not actively limit ITT
- No FADEC channel transfers during the start.

The distinction between Flight Mode and Ground Mode, for the purpose of starting, is if the Calibrated Air Speed is greater than 75 kt (139 km/h) then Flight Mode is set. In fault conditions when CAS is not available, the WOW discrete from the Propeller Electronic Control (PEC) is used to determine Ground/Flight Mode Status.

Ignition Control During Starts

[Refer Figure 80-4](#)

The dual channel ignition system is powered electrically by the aircraft essential buses and activated by the FADEC. The ignition system is configured such that both ignitors can be commanded by each FADEC channel, while maintaining electrical isolation between the FADEC channels and the aircraft essential buses.

The ignition selection from the flight compartment is sent to the FADEC on an ARINC 429 bus through the Engine Cockpit Interface Unit (ECIU).

The FADEC ignition system has two flight compartment selectable modes of operation:

- 1–OFF: The FADEC disables ignition regardless of ground or flight status.
- 2–NORM: FADEC activates ignition during engine starts (ground or flight starts). FADEC commands both ignitors on during flameout and surge accommodation.

In the event of loss of communication from the ECIU, by default the FADEC selects NORMAL ignition.

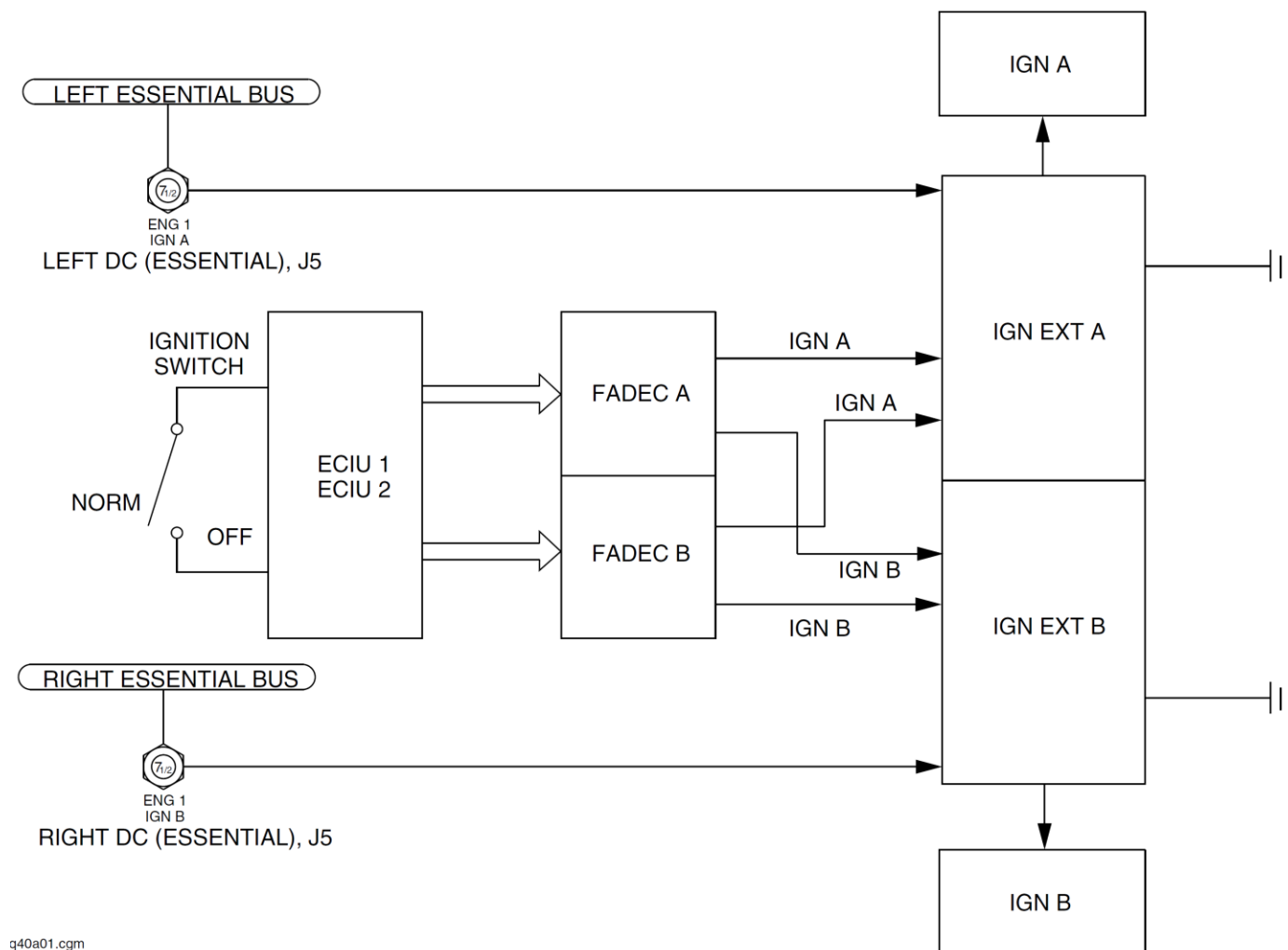


Figure 80-4 Ignition System Electrical Block Diagram

Engine Starting – Electrical Operation

Refer Figure 80-5

When powered up, the FADEC does a series of initialization tests on internal functions and interfaces. If tests are passed, the FADEC begins transmitting data on ARINC 429 and RS 422 serial data buses.

With the ignition switch set to NORM, the ignition exciter is armed and will start to fire the ignitors when it receives a signal from the FADEC that N_h is greater than 8%.

The engine control system is powered with 28 VDC from the left Essential Bus (circuit breaker H6). The power energizes pin 1 of the Engine Start Panel and pin F9 of relay K3 Relay Junction Box No. 2.

The start sequence that follows is for #1 Engine, the sequence for #2 Engine is similar.

Engine Start is initiated when the SELECT switchlight on the Engine Start Panel is Set to the 1 position. This action closes contacts 4 to 6 which supplies 28 VDC to:

- pin B3 of the START switchlight on the Engine Start Panel
- from pin 3 on the Engine Start Panel, power will flow to pin H5 of relay K1 on Relay Junction Box No. 2.

Refer Figure 80-6

At Relay Junction Box No. 2 Power will flow through the normally closed contacts C2 and C3 on relay K1 through the diode CR1 which supplies 28 VDC to:

- pin H15 on Relay Junction Box No. 2. Power will flow to pin C1 allowing the Time Delay Relay K3 to energize
- pin 4 on the Engine Start Panel energizes the hold coil on switch S1 which magnetically holds the SELECT switchlight in position until the end of the start sequence.

The energized Time Delay Relay K3 allows power to flow from pin F9 through contacts:

- B2 to B1 which sends a 28 VDC to the two GCU's, this limits the starting current to approximately 400 amps during the start cycle
- A2 to A1 through the diode CR4 which signals the EPCU of a start selection
- A2 to A1 through the diode CR4 which power will flow to pin 5 of the Engine Start Panel to illuminate the amber SELECT switchlight.

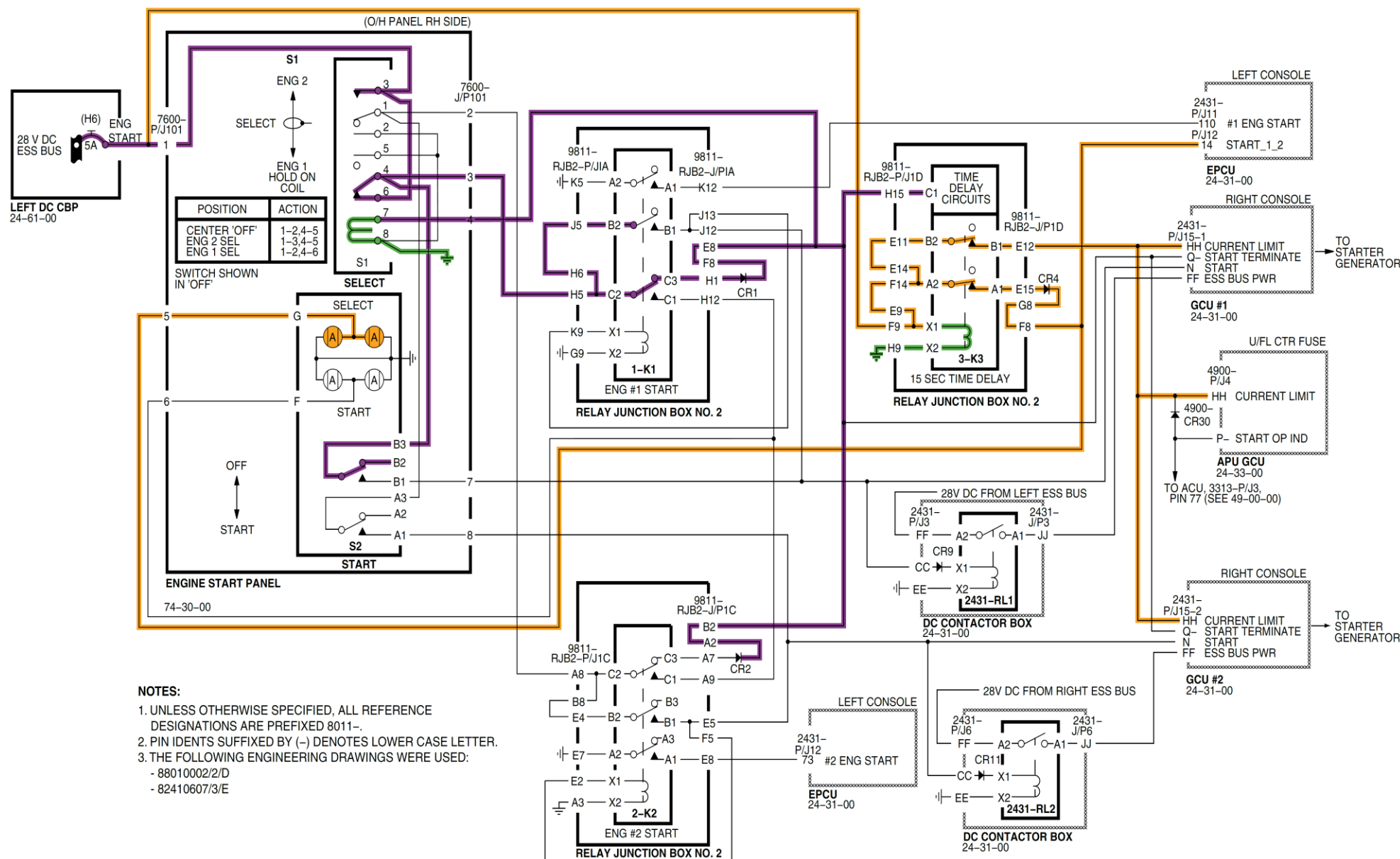


Figure 80-5 Engine No.1 Selected on Start Select Switch

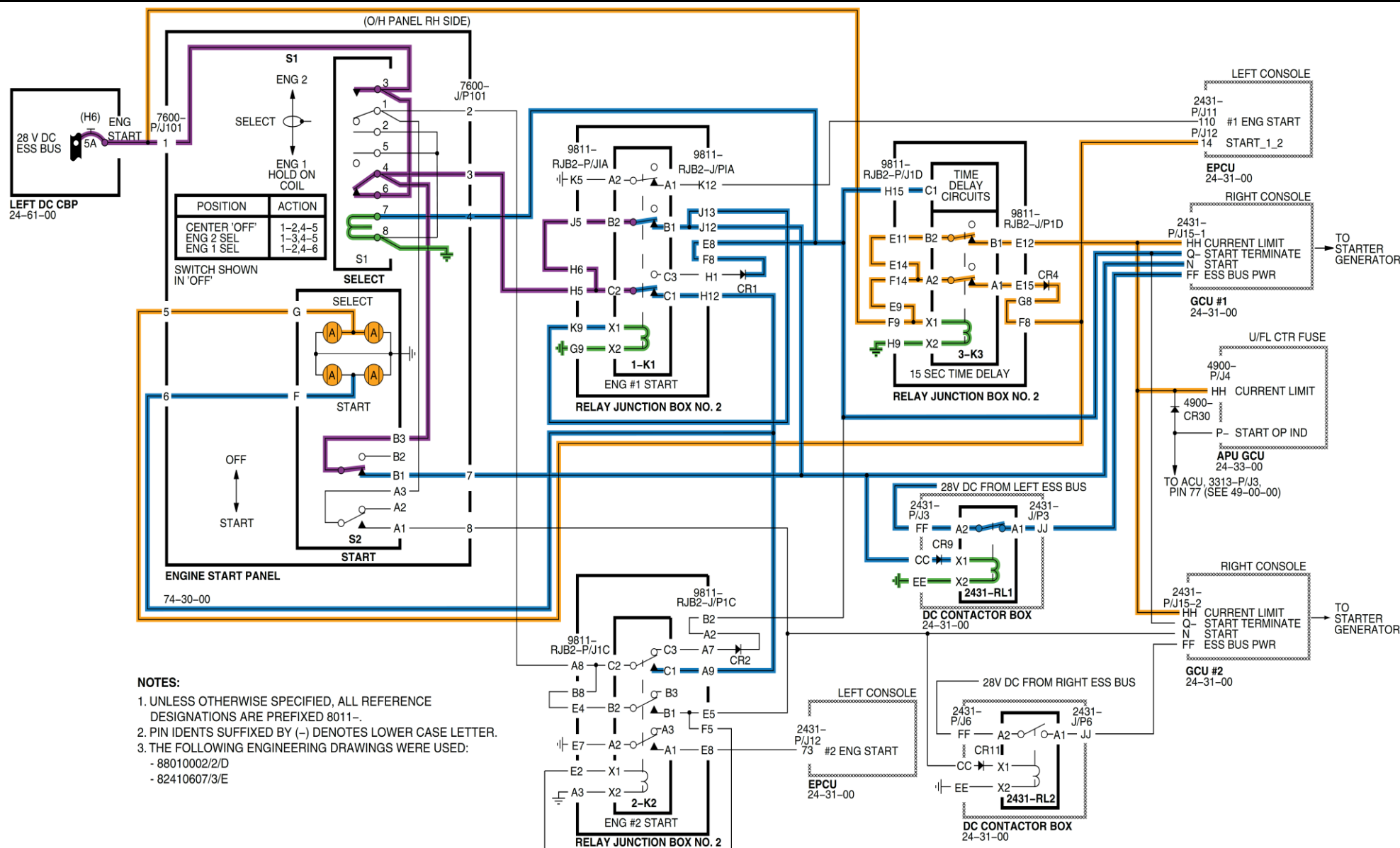


Figure 80-6 Start Switch Pushed (Momentary Action)

Refer Figure 80-6

Engine Start is activated when the START switchlight is depressed. This action closes contacts B3 to B1 and momentarily supplies 28 VDC to:

- pin J12 and J13 of the Relay Junction Box No 2 power will flow to pin K9 to energize the K1 relay
- pin CC of the DC Contactor Box that energizes relay 2431-RL1 to close contacts A2 to A1 to supply power from the left Essential Bus to drive the Starter
- pin N of the GCU 1 to command the GCU to starter mode.

The energized Engine Start Relay K1 will close contactors:

- B2 to B1 that will supply the power required to keep relay K1 energized after the START switchlight returns to the normal position
- A2 to A1 to provide a ground for the EPCU through pin 110 which verifies the status of contactor K1, when the system reverts to generator mode.
- C2 to C1 that supplies power to pin 6 on the Engine Start Panel to illuminate the amber START on the switchlight
- pin N of the GCU 1 to keep the GCU starter mode command.

When there is a rise in Nh the CLA may be positioned to START/ FEATHER. The movement of the Condition Lever from the FUEL OFF position causes:

- Condition Lever Start/Shutdown microswitches transition from the closed circuit to the open circuit state (one per channel) resetting the independent Nh Overspeed (O/S) protection circuitry to unlatch any previous O/S trip
- Wrap-around signals from the Start/Shutdown discretes are input to the FADEC controller, signalling to the software that a start has been commanded
- When Nh is greater than 8% and Condition Lever Angle (CLA) is not in the FUEL OFF position, FADEC begins metering fuel flow.

When the engine RPM is greater than 8% Nh, FADEC signals the:

- FMU to deliver metered fuel

- ignition exciter to energize the ignitor to No.1 engine.

When the engine RPM is greater than 50% Nh, the starter speed sensor signals the GCU to terminate the 28Vdc output from pin Q, that will cause:

- Power is removed from pin C1 of time delay relay K3, but because of the time delay, it retains current limiting for 15 seconds.
- Power is removed from pin 4 on the Engine Start Panel that de-energizes the hold relay of the SELECT switch and it returns to the OFF position
- This separates the contacts 4 to 6 removes power from pin 3 that de-energizes relay K1
- This separates the contacts B2 to B1 that will terminate the power to relay K1 to cause the START switchlight to go out
- This separates the contacts A2 to A1 to terminate the ground for the EPCU through pin 110
- This separates the contacts C2 to C1 that terminates power to pin 6 on the Engine Start Panel the amber START on the switchlight
- Power is removed from pin N of the GCU 1 to terminate the start mode command.

Abnormal Operations

Ignition Control During Flameout and Engine Surge

In addition to controlling the ignitors during starting, FADEC can determine that the engine has suffered a flameout within 1 second (rapid decrease of EGT and Nh).

FADEC will activate the ignition circuitry to the two ignitors as soon as the flameout has been detected and will cancel when the engine has recovered

The FADEC automatically aborts the start, and stops fuel flow if any of the following conditions occur:

- The engine does not light up within 16 seconds of fuel flow on. (FADEC starts a fault count)
- ITT limit is greater than 920°C (on the ground)
- Nh does not reach 50% within 70 seconds (i.e. hung or slow start).